MAHATMAGANDHIUNIVERSITY,KOTTAYAM
MGU-UGP (HONOURS)
FIRSTSEMESTEREXAMINATION
(2024 ADMISSION ONWARDS)

MG2MDCSTA100–TimeSeriesMethodsandtheirApplications PRACTICAL EXAMINATION

Duration:1Hour Maximum:35 Marks

Fromthefollowingeightquestions,studentsmustanswerany**five**questions.Each question carries **7** marks.

[Solve the questions using R Studio in the computer lab. The R code andoutputs (except graphs) must be written in the answer paper. Ensure that the graphs generated are evaluatedbytheinvigilatorbeforeleavingthelab.Documentyouranalysisandfindings.]

1. Createatimeseriesobjectstartingfrom Jan2020withmonthlyfrequencyandplot all the time series plots for the following time series data.

   [Analyse][CO4] 1.06, 2.63, 1.50, 0.01, -0.43, -0.38, 0.93, 1.54, -2.83, , -2.26, -2.43, -0.11, -0.88, -0.30, -3.70, -1.46, 0.12, 1.25, 0.60, 0.22, 1.54, 4.46, 3.72, 8.78, 2.30, 2.58, 4.15, 0.73, -2.63, 4.83

2. Create a time series object starting from April 2020 with monthly frequency and plot all the time series plots for the following time series data.

   [Analyse] [CO4]

   0.76, 0.74, 0.77, 0.50, -0.31, 1.63, 1.31, -1.31, 0.05, -0.45, -1.29, -0.86, -1.46, -1.46, -1.35, -2.36, -0.34, -0.02, -1.15, 0.68, 0.76, 0.09, 0.94, 1.35, 1.50

3. Create a time series object starting from June 2020 with monthly frequency and plot all the time series plots for the following time series data.

   [Analyse][CO4] 5.89, 3.32, -0.33, 3.51, 3.64, 4.70, 2.59, 6.28, 8.03, 8.18, 6.41, 3.70, 1.23, -0.84, -0.18, -1.60, 0.15, 1.25, 3.71, 3.23, 4.81, 1.59, 2.55, 2.51, 2.01, 5.05, 5.91

4. Createatimeseriesobjectstartingfrom Jan2021withmonthlyfrequencyandplot all the time series plots for the following time series data.

   [Analyse][CO4]

   -2.43, 1.09, -6.26, -12.41, -8.40, -5.17, 4.40, 2.70, 0.78, 3.90, 3.26, 10.81, 7.19,-1.63,2.29,1.89,-6.35,-13.31,-11.94, -9.61, -4.85, 1.82, -1.01,-3.14,-2.93

5. Createatimeseriesobjectstartingfrom Jan 2000 withmonthlyfrequencyandplot all the time series plots for the following time series data.

[Analyse][CO4]

48.37, 50.79, 51.83, 50.64, 48.93, 49.98, 50.41, 50.55, 51.61, 51.43, 48.39, 48.05, 50.78, 50.68, 49.25, 49.44, 49.97, 49.44, 49.42, 50.46, 52.09, 50.45, 48.22, 47.64, 51.32, 51.75, 50.64, 48.31

6. PlotallthetimeseriesplotsfortheAirPassengersdata(abuilt-intimeseriesdata set on monthly airline passenger totals from 1949 to 1960) in R.

[Analyse][CO4]

7. CreateatimeseriesobjectandfitanARmodelforthefollowingtimeseries data.

[Skill][CO4] 1.06, 2.63, 1.50, 0.01, -0.43, -0.38, 0.93, 1.54, -2.83, , -2.26, -2.43, -0.11, -0.88, -0.30, -3.70, -1.46, 0.12, 1.25, 0.60, 0.22, 1.54, 4.46, 3.72, 8.78, 2.30, 2.58, 4.15, 0.73, -2.63, 4.83

8. CreateatimeseriesobjectandfitanARmodelforthe followingtimeseries data.

[Skill][CO4] 0.76, 0.74, 0.77, 0.50, -0.31, 1.63, 1.31, -1.31, 0.05, -0.45, -1.29, -0.86, -1.46, -1.46, -1.35,-2.36, -0.34, -0.02, -1.15, 0.68, 0.76, 0.09, 0.94, 1.35, 1.50

9. CreateatimeseriesobjectandfitanARmodelforthefollowingtimeseries data.

[Skill][CO4] 5.89, 3.32, -0.33, 3.51, 3.64, 4.70, 2.59, 6.28, 8.03, 8.18, 6.41, 3.70, 1.23, -0.84, -0.18, -1.60, 0.15, 1.25, 3.71, 3.23, 4.81, 1.59, 2.55, 2.51, 2.01, 5.05, 5.91

10. CreateatimeseriesobjectandfitanARmodelforthefollowingtimeseries data.

[Skill][CO4]

-2.43, 1.09, -6.26, -12.41, -8.40, -5.17, 4.40, 2.70, 0.78, 3.90, 3.26, 10.81, 7.19,-1.63,2.29,1.89,-6.35,-13.31,-11.94, -9.61, -4.85, 1.82, -1.01,-3.14,-2.93

11. CreateatimeseriesobjectandfitanARmodelforthefollowingtimeseries data.

[Skill][CO4]

48.37, 50.79, 51.83, 50.64, 48.93, 49.98, 50.41, 50.55, 51.61, 51.43, 48.39, 48.05, 50.78, 50.68, 49.25, 49.44, 49.97, 49.44, 49.42, 50.46, 52.09, 50.45, 48.22, 47.64, 51.32, 51.75, 50.64, 48.31

12. FitanARmodelfortheAirPassengersdata(abuilt-intimeseriesdataseton monthly airline passenger totals from 1949 to 1960) in R.

[Skill][CO4]

13. FitanARmodelforthesunspot.yeardata(abuilt-intimeseriesdatasetonyearly sunspot counts from 1749 to 1983) in R. [Skill][CO4]
14. Fit an AR model for the Nile data (a built-in time series data set on annual river flow data from 1871 to 1970) in R. [Skill][CO4]
15. Fit an AR model for the LakeHuron data (a built-in time series data seton annual water levels of Lake Huron from 1875 to 1972) in R. [Skill][CO4]
16. Simulate 30 values of the specified time series model and plot all the time series plots. Also note down the first 5 and last 5 values generated.

[Interest][CO4]

- AR(1)modelusing φ_1=0.70andseedvalue 13
- AR(2)modelusingφ_1=0.50,φ_2=0.33,intercept55andseedvalue7

17. Simulate 30 values of the specified time series model and plot all the time series plots. Also note down the first 5 and last 5 values generated.

[Interest][CO4]

- AR(1)modelusing φ_1=-0.20,intercept75andseedvalue99
- AR(2)modelusingφ_1=0.60,φ_2=0.20andseedvalue345

18. Simulate 30 values of the specified time series model and plot all the time series plots. Also note down the first 5 and last 5 values generated.

[Interest][CO4]

- AR(1)modelusingφ_1=0.80,intercept50andseedvalue8
- AR(2)modelusingφ_1=1.50,φ_2=-0.75andseedvalue888

19. Simulate 30 values of the specified time series model and plot all the time series plots. Also note down the first 5 and last 5 values generated.

[Interest][CO4]

- AR(1)modelusing φ_1=0.70andseedvalue 13
- AR(2)modelusingφ_1=0.60,φ_2=0.20,intercept55andseedvalue345

20. Simulate 30 values of the specified time series model and plot all the time series plots. Also note down the first 5 and last 5 values generated.

[Interest][CO4]

- AR(1)modelusingφ_1=-0.20andseedvalue99

- AR(2)modelusingφ_1=1.50, φ_2=-0.75,intercept50andseedvalue888

21. Simulate 30 values of the specified time series model and plot all the time series plots. Also note down the first 5 and last 5 values generated.
[Interest][CO4]
    - AR(1)modelusingφ_1=0.80,intercept60andseedvalue8
    - AR(2)modelusingφ_1=0.50,φ_2=0.33andseedvalue7
22. Simulate 30 values of the specified time series model and plot all the time series plots. Also note down the first 5 and last 5 values generated.
[Interest][CO4]
    - AR(1)modelusingφ_1=0.70,intercept55andseedvalue13
    - AR(2)modelusingφ_1=1.50,φ_2=-0.75andseedvalue888
23. Simulate 30 values of the specified time series model and plot all the time series plots. Also note down the first 5 and last 5 values generated.
[Interest][CO4]
    - AR(1)modelusingφ_1=-0.20andseedvalue99
    - AR(2)modelusingφ_1=0.50,φ_2=0.33,intercept45andseedvalue7
24. Simulate 30 values of the specified time series model and plot all the time series plots. Also note down the first 5 and last 5 values generated.
[Interest][CO4]
    - AR(1)modelusing φ_1=0.80andseedvalue 8
    - AR(2)modelusingφ_1=0.60,φ_2=0.20,intercept40andseedvalue345
25. Simulate 30 values of the specified time series model and plot all the time series plots. Also note down the first 5 and last 5 values generated.
[Interest][CO4]
    - AR(1)modelusingφ_1=0.70,intercept50andseedvalue13
    - AR(2)modelusingφ_1=0.50,φ_2=0.33andseedvalue7
26. Simulate 30 values of the specified time series model and plot all the time series plots. Also note down the first 5 and last 5 values generated.
[Interest][CO4]
    - AR(1)modelusingφ_1=-0.20andseedvalue99
    - AR(2)modelusingφ_1=0.60,φ_2=0.20,intercept50andseedvalue345
27. Simulate 30 values of the specified time series model and plot all the time series plots. Also note down the first 5 and last 5 values generated.
[Interest][CO4]
    - AR(1)modelusing φ_1=0.80andseedvalue 8
    - AR(2)modelusingφ_1=1.50,φ_2=-0.75,intercept100andseedvalue888
28. Simulate 30 values of the specified time series model and plot all the time series plots. Also note down the first 5 and last 5 values generated.
[Interest][CO4]

- AR(1)modelusingφ_1=0.70,intercept125andseedvalue13
- AR(2)modelusingφ_1=0.60,φ_2=0.20andseedvalue345

29. Simulate 30 values of the specified time series model and plot all the time series plots. Also note down the first 5 and last 5 values generated. [Interest][CO4]
    - AR(1)modelusing φ_1=-0.20,intercept55andseedvalue99
    - AR(2)modelusingφ_1=1.50,φ_2=-0.75andseedvalue888
30. Simulate 30 values of the specified time series model and plot all the time series plots. Also note down the first 5 and last 5 values generated. [Interest][CO4]
    - AR(1)modelusing φ_1=0.80andseedvalue 8
    - AR(2)modelusingφ_1=0.50,φ_2=0.33,intercept75andseedvalue7
31. Simulate 30 values of the specified time series model and plot all the time series plots. Also note down the first 5 and last 5 values generated. [Interest][CO4]
    - AR(1)modelusing φ_1=0.70andseedvalue 13
    - AR(2)modelusingφ_1=1.50, φ_2=-0.75,intercept25andseedvalue888
32. Simulate 30 values of the specified time series model and plot all the time series plots. Also note down the first 5 and last 5 values generated. [Interest][CO4]
    - AR(1)modelusing φ_1=-0.20,intercept10andseedvalue99
    - AR(2)modelusingφ_1=0.50,φ_2=0.33andseedvalue7
33. Simulate 30 values of the specified time series model and plot all the time series plots. Also note down the first 5 and last 5 values generated. [Interest][CO4]
    - AR(1)modelusingφ_1=0.80,intercept50andseedvalue8
    - AR(2)modelusingφ_1=0.60,φ_2=0.20andseedvalue345
34. Simulate 15 white noise values using seed values 1 and 100 and plot them. Also note down all the 15 values generated. [Skill][CO4]
35. Simulate15whitenoisevaluesusingseedvalues234and143andplotthem.Also note down all the 15 values generated. [Skill][CO4]
36. Simulate 15 white noise values using seed values 888 and 88 and plot them. Also note down all the 15 values generated. [Skill][CO4]
37. Simulate15whitenoisevaluesusingseedvalues7and18andplotthem.Alsonote down all the 15 values generated. [Skill][CO4]
38. Simulate 15 white noise values using seed values 45 and 99 and plot them. Also note down all the 15 values generated. [Skill][CO4]

39. Simulate 15 white noise values using seed values 16325 and 88 and plot them.Also note down all the 15 values generated. [Skill][CO4]
40. Simulate 15 white noise values using seed values 888 and 5433 and plot them.Alsonotedownallthe 15values generated. [Skill][CO4]

(5x7=35)